



# Methods

## Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- Articulatory matrices (ultrasound/MRI):
  - The temporal dimension of the video clip for each token was normalized to 5 frames (FR-MRI/AE-MRI) & 10 frames (BE-US).
  - Raw video → 3D CNN + MLP → 1024D latent matrix



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## Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- Trains the models to find the 1024 dimensions that best capture the information that could be used for data reconstruction.